

Supplementary Material: Complete Development Results and Historical Analyses

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TABLE I

NOMINAL 95% INTERVALS ON THE LEGACY-ARRAY FOLDS (THREE FOLDS, 68 TEST WINDOWS EACH). COVERAGE IS THE SHARE OF DISTRICT-WEEK TARGETS INSIDE THE INTERVAL; WIDTH IS THE MEAN UPPER MINUS LOWER BOUND IN CASES. THE FIRST TWO ROWS NEED NO NETWORK.

Interval	Coverage (%)	Width
Persistence, residual quantiles	96.0	99
Persistence, scaled residual quantiles	96.3	74
STGAT, Gaussian head	95.6	123
A3TGCN, Gaussian head	96.2	230
ASTGCN, Gaussian head	94.7	128
DCRNN, Gaussian head	95.2	250
AAGCN, Gaussian head	95.9	145

I. SCOPE

These materials preserve historical analyses omitted from the six-page manuscript. They are development or legacy evidence, not results of a corrected final or untouched-period evaluation. The supplementary file is provided for author review; its eligibility for submission requires a venue decision.

II. LEGACY INTERVAL COMPARISON

The Gaussian-head results exist only on legacy-array folds containing the source-table error. The persistence interval uses per-horizon training-residual quantiles; the scaled version divides residuals by the square root of one plus the latest count, then reverses that scaling for prediction. Lower bounds are clipped at zero. Coverage and width are averaged across folds, and no new model is trained for this comparison. These historical intervals are not a calibration contribution of the rebuilt-data architecture.

III. STORED DEVELOPMENT PROTOCOLS

TABLE II

STORED REBUILT-DATA DEVELOPMENT PROTOCOLS. EACH NEURAL ARM USES THREE SEEDS; PERSISTENCE IS DETERMINISTIC. BOTH USE 30 VALIDATION FORECAST STARTS. TEST SHARES DESCRIBE FORECAST STARTS, WHOSE TARGET WEEKS OVERLAP PARTITION BOUNDARIES.

Protocol	Origin fractions	Test share
Nine-origin	0.50–0.90, step 0.05	5%
Three-origin	0.55, 0.70, 0.85	15%

IV. HISTORICAL THREE-ORIGIN ABLATIONS

Table III shows the three-origin ablation, which we read as indicative only. With three origins no gap can reach p below 0.25. The simple graph control, a GCN with a residual head, scores 34.50 against 36.02 for persistence. Removing the graph gives 34.57 and a learned adjacency gives 34.17, so the whole spread across the three graph choices is 0.41 cases, well inside the standard deviation of 10 to 11 over runs. Under the shared fixed training settings, direct heads have high error on two adapted encoders: STGAT scores 62.00 and DCRNN scores 73.52 against persistence at 36.02. Their gated SEIR versions score 35.84 and 35.52.

To separate the anchor from the physics, we ran a gated head without the simulator. On validation, the simulator has mixed effects for the three poorly fitted direct-head models: the SEIR head minus the gated head is +0.66 (STGAT), 0.00 (A3T-GCN) and −0.68 (DCRNN) in RMSE. The anchor and gate account for 1.12, 1.00 and 0.96 of the repair. On test, the SEIR head is lower than the gated head by 0.62, 1.66 and 3.50. Validation and test disagree. These mixed results do not isolate an accuracy gain from the simulator. Equal validation-only tuning and matched adaptive-encoder controls are missing.

Among the displayed head controls, the lowest three-origin error is from AAGCN with a gated head, at 33.53 against 36.02 for persistence. This gap is untested. The nine-origin protocol contains no gated head without physics, and its ASTGCN arm with a residual head is 0.11 above persistence.

V. SEED VARIATION AND NORMALIZED SKILL

Table IV separates initialization variation from the between-origin variation underlying the paired intervals in the manuscript. The CSV summary records all stored arms, including horizon errors, MAE, skill, paired intervals, both BH and Holm corrections, and mean within-origin seed standard deviation. The displayed subset follows the main manuscript; no claim of significance is inferred from a positive skill.

TABLE III

HISTORICAL DEVELOPMENT ABLATION ON THE THREE-ORIGIN PROTOCOL (REBUILT DATA, ORIGINS 0.55, 0.70, 0.85, 3 SEEDS, 9 NEURAL RUNS PER ROW). WITH THREE ORIGINS AN EXACT SIGN-FLIP TEST CANNOT RETURN $p < 0.25$, SO NONE OF THESE GAPS IS TESTED. RMSE IN CASES PER DISTRICT-WEEK; $\pm$ IS THE STANDARD DEVIATION OVER THE NINE RUNS. Δ IS TEST RMSE MINUS THE GCN CONTROL (GRAPH LAYER, RESIDUAL HEAD), AVERAGED OVER THE NINE MATCHED RUNS; WINS COUNTS RUNS OUT OF NINE. DO NOT COMPARE THESE NUMBERS WITH THE NINE-ORIGIN TABLE IN THE MAIN MANUSCRIPT: THE TEST WEEKS DIFFER.

Model	Val	$h=1$	$h=2$	$h=3$	Test RMSE	Δ	Wins
Persistence (last week)	17.86	29.62	35.13	42.11	36.02 ± 12.41	–	–
GCN, residual (baseline GNN)	16.84	28.28	34.01	40.13	34.50 ± 10.41	–	–
No graph, residual	16.90	28.54	34.05	40.08	34.57 ± 11.09	+0.07	6/9
Adaptive GCN (simple), residual	16.69	28.44	33.97	39.20	34.17 ± 10.72	–0.33	7/9
AAGCN, direct	16.71	29.34	34.79	40.36	35.15 ± 11.53	+0.65	4/9
AAGCN, gated	16.57	27.78	32.89	38.93	33.53 ± 10.14	–0.97	9/9
AAGCN, SEIR, free rate	24.01	35.04	50.29	59.92	49.54 ± 18.49	+15.04	0/9
AAGCN, gated SEIR, free rate	17.43	28.68	35.33	41.46	35.58 ± 11.50	+1.08	3/9
STGAT, direct	22.93	60.24	61.97	63.67	62.00 ± 25.54	+27.50	0/9
STGAT, gated SEIR, free rate	17.63	28.34	35.30	42.39	35.84 ± 11.65	+1.34	3/9
DCRNN, direct	34.64	73.19	73.53	73.84	73.52 ± 32.90	+39.02	0/9
DCRNN, gated SEIR, free rate	17.67	28.34	34.94	41.88	35.52 ± 11.42	+1.02	3/9

TABLE IV

ADDITIONAL DESCRIPTIVE STATISTICS FOR THE HISTORICAL DEVELOPMENT COMPARISON. SKILL IS THE MEAN OVER ORIGINS OF ONE MINUS MODEL RMSE DIVIDED BY PERSISTENCE RMSE. SEED SD IS THE MEAN WITHIN-ORIGIN STANDARD DEVIATION OVER NEURAL SEEDS; PERSISTENCE IS DETERMINISTIC. NEITHER STATISTIC IS A SIGNIFICANCE TEST.

Model	Skill	Seed SD
Persistence (last week)	+0.000	0.000
GCN, direct	+0.019	0.617
GCN, residual (baseline GNN)	+0.019	0.507
LSTM, direct	+0.014	0.412
ASTGCN, direct	–0.012	0.912
AAGCN, direct	–0.007	1.429
STGAT, direct	–0.600	4.733
ASTGCN, residual	+0.009	0.536
Adaptive GCN, residual	+0.028	0.743
GCN, SEIR, free rate	–0.404	0.171
GCN, gated SEIR, free rate	+0.008	0.190
Adaptive GCN, SEIR, anchored	–0.027	0.843
Adaptive GCN, gated SEIR, mass-action	+0.008	0.098
GCN, gated SEIR, anchored	+0.033	0.292
Adaptive GCN, gated SEIR, anchored	+0.040	0.419
Adaptive GCN, gated SEIR, anchored, learned E_0	+0.035	0.448

VI. COMPLETE NINE-ORIGIN MODEL RESULTS

Table V reports every learned configuration from the nine-origin development evaluation.

TABLE V

COMPLETE NINE-ORIGIN DEVELOPMENT RESULTS. Δ IS THE MEAN PAIRED RMSE DIFFERENCE FROM PERSISTENCE; CI IS THE DESCRIPTIVE 95% t INTERVAL OVER NINE SEED-AVERAGED ORIGINS. p_{raw} IS THE EXACT TWO-SIDED SIGN-FLIP VALUE AND p_{Holm} IS ADJUSTED ACROSS THE 30 LEARNED CONFIGURATIONS.

Model	RMSE	MAE	Δ	95% CI	p_{raw}	p_{Holm}
Persistence (last week)	28.54	13.94	–	–	–	–
GCN, direct	28.17	13.66	-0.37	[-1.88, +1.14]	0.594	1.000
GCN, residual	28.12	13.62	-0.42	[-1.54, +0.70]	0.391	1.000
GCN, gated SEIR, free rate	28.43	13.51	-0.11	[-3.23, +3.00]	0.910	1.000
GCN, SEIR, free rate	40.77	17.97	+12.23	[+2.67, +21.80]	0.008	0.227
LSTM, direct	28.73	14.00	+0.19	[-1.99, +2.37]	0.887	1.000
LSTM, residual	27.73	13.43	-0.81	[-2.31, +0.68]	0.219	1.000
LSTM, gated SEIR, free rate	28.47	13.53	-0.07	[-3.29, +3.14]	0.957	1.000
LSTM, SEIR, free rate	40.58	17.97	+12.04	[+2.45, +21.63]	0.008	0.227
STGAT, direct	44.71	22.17	+16.17	[+4.24, +28.11]	0.008	0.227
STGAT, residual	28.73	14.04	+0.19	[-0.35, +0.74]	0.777	1.000
STGAT, gated SEIR, free rate	28.60	13.54	+0.06	[-3.37, +3.48]	0.969	1.000
STGAT, SEIR, free rate	44.67	20.38	+16.13	[+3.53, +28.73]	0.008	0.227
A3T-GCN, direct	48.37	24.27	+19.83	[+4.66, +35.00]	0.008	0.227
A3T-GCN, residual	28.24	13.76	-0.30	[-1.52, +0.92]	0.523	1.000
A3T-GCN, gated SEIR, free rate	28.47	13.52	-0.07	[-3.27, +3.13]	0.949	1.000
A3T-GCN, SEIR, free rate	42.26	19.18	+13.71	[+3.38, +24.05]	0.004	0.117
ASTGCN, direct	28.94	14.27	+0.40	[-2.02, +2.81]	0.750	1.000
ASTGCN, residual	28.65	13.98	+0.11	[-1.68, +1.90]	0.902	1.000
ASTGCN, gated SEIR, free rate	28.53	13.57	-0.01	[-3.24, +3.21]	0.984	1.000
ASTGCN, SEIR, free rate	41.92	18.14	+13.37	[+1.88, +24.87]	0.012	0.270
AAGCN, direct	29.26	14.06	+0.72	[-1.57, +3.00]	0.547	1.000
AAGCN, residual	28.61	13.87	+0.07	[-2.09, +2.22]	0.977	1.000
AAGCN, gated SEIR, free rate	28.49	13.50	-0.05	[-3.28, +3.17]	0.961	1.000
AAGCN, SEIR, free rate	41.50	18.58	+12.96	[+2.94, +22.97]	0.008	0.227
Adaptive GCN, residual	27.94	13.61	-0.60	[-1.81, +0.61]	0.301	1.000
Adaptive GCN, gated SEIR, anchored	27.45	13.31	-1.09	[-2.81, +0.63]	0.188	1.000
Adaptive GCN, gated SEIR, anchored, learned E_0	27.45	13.29	-1.10	[-2.84, +0.65]	0.199	1.000
Adaptive GCN, SEIR, anchored	29.80	13.84	+1.26	[-1.47, +3.98]	0.383	1.000
Adaptive GCN, gated SEIR, mass-action	28.39	13.52	-0.16	[-3.16, +2.85]	0.887	1.000
GCN, gated SEIR, anchored	27.62	13.39	-0.92	[-2.60, +0.76]	0.230	1.000